

Mini Project-

E-Commerce Order Analytics System

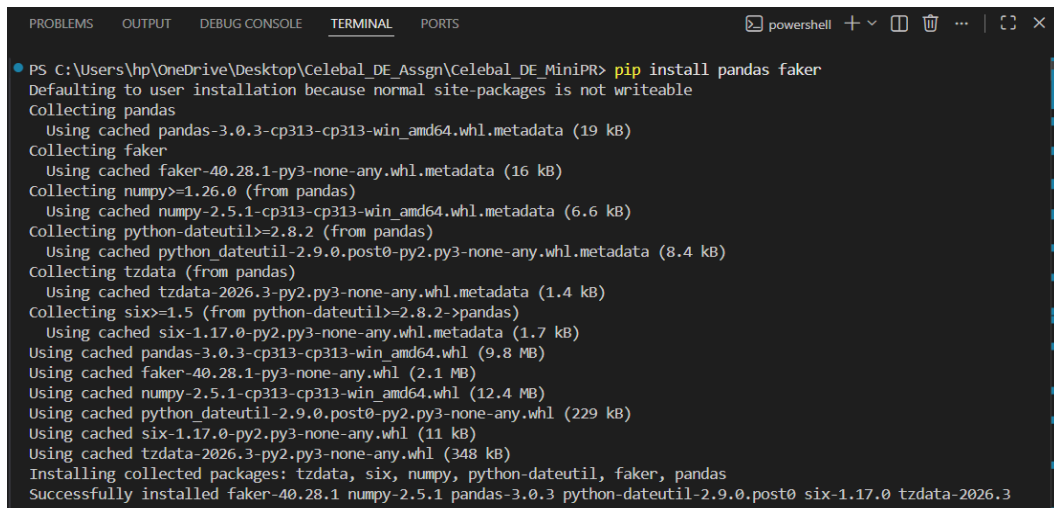
Objective

Build a Python command-line application that:

- Accepts user arguments
- Connects to SQLite
- Runs SQL queries dynamically
- Displays results in a formatted table
- Handles invalid input
- Handles empty results
- Handles database connection errors

STEP 1 Install tabulate

I. Open VS Code Terminal and Import libraries:



```
PS C:\Users\hp\OneDrive\Desktop\Celebal DE Assgn\Celebal DE MiniPR> pip install pandas faker
Defaulting to user installation because normal site-packages is not writeable
Collecting pandas
  Using cached pandas-3.0.3-cp313-cp313-win_amd64.whl.metadata (19 kB)
Collecting faker
  Using cached faker-40.28.1-py3-none-any.whl.metadata (16 kB)
Collecting numpy>=1.26.0 (from pandas)
  Using cached numpy-2.5.1-cp313-cp313-win_amd64.whl.metadata (6.6 kB)
Collecting python-dateutil>=2.8.2 (from pandas)
  Using cached python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting tzdata (from pandas)
  Using cached tzdata-2026.3-py2.py3-none-any.whl.metadata (1.4 kB)
Collecting six>=1.5 (from python-dateutil>=2.8.2->pandas)
  Using cached six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Using cached pandas-3.0.3-cp313-cp313-win_amd64.whl (9.8 MB)
Using cached faker-40.28.1-py3-none-any.whl (2.1 MB)
Using cached numpy-2.5.1-cp313-cp313-win_amd64.whl (12.4 MB)
Using cached python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Using cached tzdata-2026.3-py2.py3-none-any.whl (348 kB)
Installing collected packages: tzdata, six, numpy, python-dateutil, faker, pandas
Successfully installed faker-40.28.1 numpy-2.5.1 pandas-3.0.3 python-dateutil-2.9.0.post0 six-1.17.0 tzdata-2026.3
```

II. Run: pip install tabulate

Verify: pip show tabulate

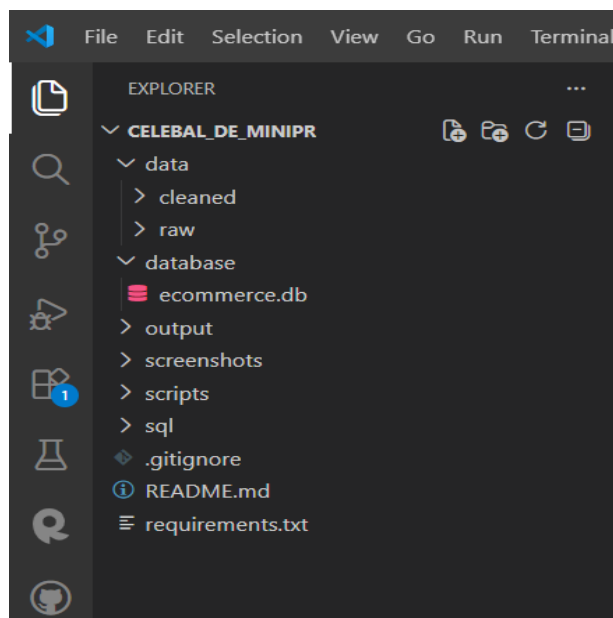
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\hp\OneDrive\Desktop\Celebal_DE_Assgn\Celebal_DE_MinipR> pip install tabulate
Defaulting to user installation because normal site-packages is not writeable
Collecting tabulate
  Using cached tabulate-0.10.0-py3-none-any.whl.metadata (40 kB)
Using cached tabulate-0.10.0-py3-none-any.whl (39 kB)
Installing collected packages: tabulate
Successfully installed tabulate-0.10.0

[notice] A new release of pip is available: 25.1 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\hp\OneDrive\Desktop\Celebal_DE_Assgn\Celebal_DE_MinipR> pip show tabulate
Name: tabulate
Version: 0.10.0
Summary: Pretty-print tabular data
Home-page: https://github.com/astanin/python-tabulate
Author:
Author-email: Sergey Astanin <s.astanin@gmail.com>
License-Expression: MIT
Location: C:\Users\hp\AppData\Roaming\Python\Python313\site-packages
Requires:
Required-by:

```

Project Structure:



STEP 2: Report 1 – Dataset Summary

Raw Dataset

Dataset	Records
Orders	700
Customers	500
Products	500

Dataset	Records
Orders	700
Order Items	1500

Cleaned Dataset

Dataset Records

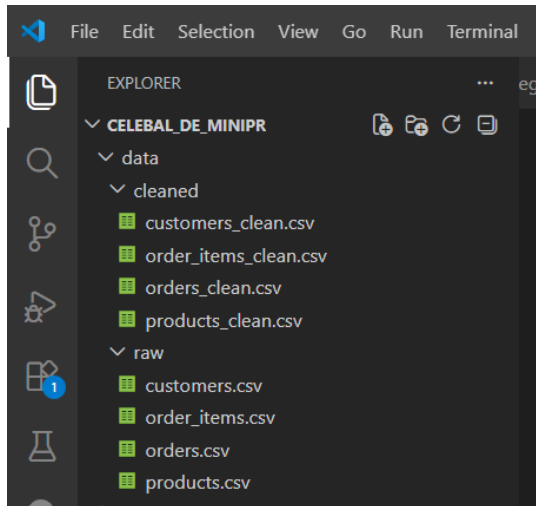
Customers 492

Products 498

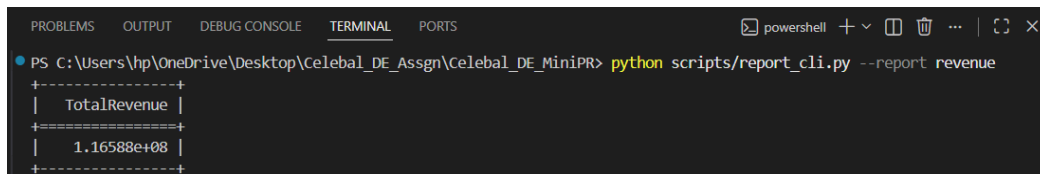
Dataset Records

Orders 665

Order Items 1432



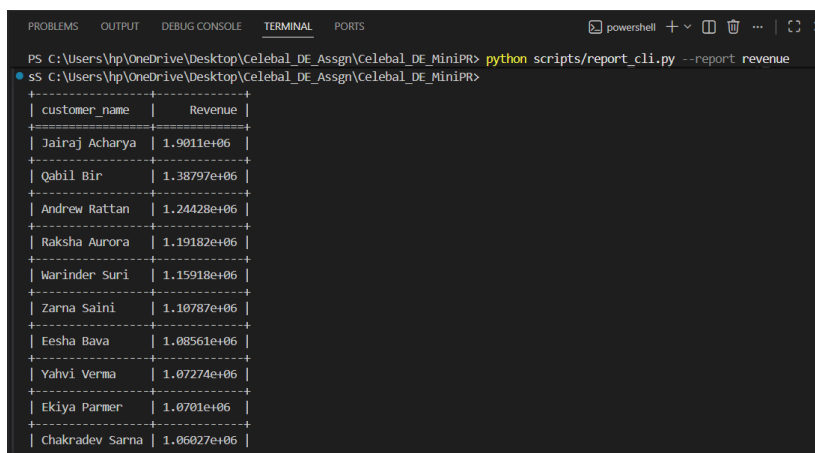
Report 2 – Total Revenue



Report 3 – Top 10 Customers

SQL Query

Revenue per customer.



Report 4 – Revenue by Category

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell + v [] [] ...
PS C:\Users\hp\OneDrive\Desktop\Celebal_DE_Assgn\Celebal_DE_MiniPR> python scripts/revenue_by_category.py

Revenue By Category

+-----+-----+
| category | Revenue |
+=====+=====+
| Clothing | 3.40235e+07 |
+-----+-----+
| Electronics | 3.15989e+07 |
+-----+-----+
| Books | 2.68336e+07 |
+-----+-----+
| Home | 2.41321e+07 |
+-----+-----+
```

Report 5 – Monthly Revenue Trend

```
PS C:\Users\hp\OneDrive\Desktop\Celebal_DE_Assgn\Celebal_DE_MiniPR> python scripts/monthly_revenue.py

Monthly Revenue

+-----+-----+
| Month | Revenue |
+=====+=====+
| 2024-07 | 2.55908e+06 |
+-----+-----+
| 2024-08 | 5.10102e+06 |
+-----+-----+
| 2024-09 | 4.73748e+06 |
+-----+-----+
| 2024-10 | 4.12832e+06 |
+-----+-----+
| 2024-11 | 3.77239e+06 |
+-----+-----+
| 2024-12 | 2.89044e+06 |
+-----+-----+
```

Report 6 – Window Functions

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell + v [] [] ...
PS C:\Users\hp\OneDrive\Desktop\Celebal_DE_Assgn\Celebal_DE_MiniPR> python scripts/window_report.py

Customer Ranking

+-----+-----+-----+
| customer_name | Revenue | Rank |
+=====+=====+=====+
| Jairaj Acharya | 1.9011e+06 | 1 |
+-----+-----+-----+
| Qabil Bir | 1.38797e+06 | 2 |
+-----+-----+-----+
| Andrew Rattan | 1.24428e+06 | 3 |
+-----+-----+-----+
| Raksha Aurora | 1.19182e+06 | 4 |
+-----+-----+-----+
| Warinder Suri | 1.15918e+06 | 5 |
+-----+-----+-----+
| Zarna Saini | 1.10787e+06 | 6 |
+-----+-----+-----+

Ln 57, Col 13
```

Report 7 – Cohort Analysis

```
PS C:\Users\hp\OneDrive\Desktop\Celebal_DE_Assgn\Celebal_DE_MinipR> python scripts/cohort_report.py
```

Cohort Analysis

CohortMonth	PurchaseMonth	Customers
2024-07	2024-07	21
2024-07	2024-08	2
2024-07	2024-09	1
2024-07	2024-11	2
2024-07	2024-12	1
2024-07	2025-01	2

Report 8 – Customer Segmentation

```
PS C:\Users\hp\OneDrive\Desktop\Celebal_DE_Assgn\Celebal_DE_MinipR> python scripts/run_segmentation.py
```

Purchase Frequency

customer_id	Orders	Segment
0	C0002	1 One-Time
1	C0003	1 One-Time
2	C0004	2 Occasional
3	C0006	2 Occasional
4	C0007	1 One-Time
5	C0008	1 One-Time
6	C0009	2 Occasional
7	C0010	2 Occasional
8	C0013	4 Occasional
9	C0014	1 One-Time
10	C0015	3 Occasional
11	C0016	4 Occasional

RFM :

```
PS C:\Users\hp\OneDrive\Desktop\Celebal_DE_Assgn\Celebal_DE_MinipR> python scripts/rfm_report.py
```

RFM Analysis

customer_id	Recency	Frequency	Monetary
C0259	273	5	1.9011e+06
C0472	399	5	1.38797e+06
C0361	674	2	1.24428e+06
C0232	167	4	1.19182e+06
C0451	527	3	1.15918e+06
C0357	325	3	1.10787e+06
C0166	188	2	1.08561e+06
C0440	85	2	1.07274e+06

Report 9 – CLI Reports

Revenue:

```
PS C:\Users\hp\OneDrive\Desktop\Celebal_DE_Assgn\Celebal_DE_MinipR> python scripts/report_cli.py --report revenue
```

TotalRevenue
1.16588e+08

Top Customers:

```
PS C:\Users\hp\OneDrive\Desktop\Celebal_DE_Assgn\Celebal_DE_MinipR> python scripts/report_cli.py --report top_customer
```

customer_name	Revenue
Jairaj Acharya	1.9011e+06
Qabil Bir	1.38797e+06
Andrew Rattan	1.24428e+06
Raksha Aurora	1.19182e+06
Warinder Suri	1.15918e+06
Zarna Saini	1.10787e+06
Eesha Bava	1.08561e+06
Yahvi Verma	1.07274e+06
Ekiya Parmer	1.0701e+06

Retention:

```
PS C:\Users\hp\OneDrive\Desktop\Celebal_DE_Assgn\Celebal_DE_MinipR> python scripts/report_cli.py --report retention
```

Cohort	Customers
2024-07	21
2024-08	23
2024-09	16
2024-10	21
2024-11	17
2024-12	15
2025-01	25
2025-02	19
2025-03	16
2025-04	15

RFM:

```
PS C:\Users\hp\OneDrive\Desktop\Celebal_DE_Assgn\Celebal_DE_MinipR> python scripts/report_cli.py --report rfm
```

customer_id	Recency	Frequency	Monetary
C0002	111	1	165820
C0003	552	1	266749
C0004	260	2	434222
C0006	369	2	226117
C0007	559	1	496732
C0008	144	1	294211
C0009	538	1	193744
C0010	425	2	681442
C0013	283	3	801563
C0014	512	1	459975

Report 10 – Edge Case Handling

```
PS C:\Users\hp\OneDrive\Desktop\Celebal_DE_Assgn\Celebal_DE_MinipR> python scripts/report_cli.py --report xyz
```

Invalid Report Name

Available Reports

```
-----
revenue
top_customers
retention
rfm
```

Missing Database:

Temporarily rename: database/ecommerce.db to database/ecommerce_backup.db

The screenshot shows a VS Code editor with a SQLite database view of 'ecommerce_backup.db'. The database contains a table named 'customers' with 491 rows. The columns are customer_id, customer_name, email, registration_date, and status. The terminal window at the bottom shows the command 'python scripts/report_cli.py --report revenue' and the error message 'Database not found.'.

customer_id	customer_name	email	registration_date	status
C0001	Shravya Chawla	vyogi@example.com	2025-01-07	VII
C0002	Aarnav Bahri	gpatil@example.net	2026-06-26	PR
C0003	Tripti Jani	charvi96@example.net	2026-02-01	PR
C0004	Yug Radhakrishnan	mitalikhanna@example.org	2023-09-23	VII
C0005	Ekbal Dhingra	chaudhurivedhika@example.com	2024-02-21	RE
C0006	Frado Chana	nahuja@example.org	2025-08-06	RE
C0007	Timothy Bhattacharyya	lzachariah@example.net	2025-03-15	RE
C0008	Imaran Kanda	deojyoti@example.net	2025-11-26	VII
C0009	Yoshita Nagi	theodore03@example.org	2024-05-05	PR
C0010	Kamya Reddy	triyabahl@example.org	2023-12-28	PR
C0011	Chatura Narang	owenpai@example.com	2026-06-19	PR
C0012	Veda Krishnamurthy	jagdishbajwa@example.com	2024-12-03	PR
C0013	Isha Bhat	niteshsubramanian@example.net	2024-02-17	PR
C0014	Harita Sengupta	zmadan@example.com	2024-06-08	PR
C0015	Vedika Dube	priya01@example.org	2026-05-24	VII
C0016	Zaid Kar	madhavadugar@example.com	2024-11-14	PR
C0017	Falak Muni	deepvictor@example.org	2025-04-25	RE
C0018	Ayush Madan	kavya38@example.com	2024-01-22	VII

```
PS C:\Users\hp\OneDrive\Desktop\Celebal_DE_Assgn\Celebal_DE_MinipR> python scripts/report_cli.py --report revenue
Database not found.
```